Problem 2

May 2, 2025

[40]:
```
import numpy as np, math as math
from decimal import Decimal, getcontext
```

(a)

[2]:
```
x = 10**8
y = 1.0

z_128 = (np.longdouble(x) + np.longdouble(y)) - np.longdouble(x)
z_64 = (np.double(x) + np.double(y)) - np.double(x)
z_32 = (np.single(x) + np.single(y)) - np.single(x)
print(z_128, z_64, z_32)
```

```
1.0 1.0 0.0
```

The expected result here is z = y = 1. Addition is not commutative in finite-precision binary arithmetic. Single precision floats are only accurate to the 7th digit, and as (x + y) is 10ˆ8 +1 = 100000001, the last digit falls outside the accuracy range. The result is that the calculation becomes (x + y) = x -> x - x = 0. This is not an issue for double precision +, as these data types use more bits and have a decimal accuracy which captures the addition of y to x.

(b)

[6]:
```
a, b = 0.1 + 0.2, 0.3
print(a == b, a - b < 1e-16)
```

```
False True
```

The reason these values don't add up is because none of the numbers can be represented exactly in e.g. double precision floats. For a decimal fraction to be exactly representable in binary, it must be expressible as p/q, where both p and q are integers composed solely of powers of 2. Numbers like 0.1, 0.2, and 0.3 can't be expressed this way, so they are stored as approximations. To circumvent this issue, an epsilon greater than the precision of the data types which are compared can be used, as seen in the code snippet.

(c)

[22]:
```
x, y, delta = 1.0, 1 + 1e6 * 1e-16, 1e-16
for _ in range(1_000_000): x += delta
print(x==y)
```

```
False
```

What's happening here is that 10ˆ-16 is on the limit of double float precision. Similarly to (a), when it it added to a (comparatively) large number such as x = 1.0, the spacing of decimal places makes it so it falls *just* outside the machine precision. Therefore, when adding small numbers to larger numbers, summing instances of small numbers first is good practice, and in this case solves the problem:

[23]:
```
delta, tot = 1e-16, 0.0
for _ in range(1_000_000): tot += delta
x = 1.0 + tot
print(x == y)
```

```
True
```

(d)

[ ]:
```
f_170, f_171 = np.double(1), np.double(1)
for n in range(1, 171): f_170 *= n
for n in range(1, 172): f_171 *= n
print(f_170, f_171)
print(f'Max representable val: {np.finfo(np.double).max}')
```

```
7.257415615307994e+306 inf
Max representable val: 1.7976931348623157e+308
```

The maximum stored value with double precision floating point is roughly 1e+308. As seen in the printout, 170! comes very close to this value, while 171! causes data overflow because it exceeds it. One strategy to fix this is to manually set a decimal precision to avoid overflow:

[49]:
```
getcontext().prec = 310
f_170, f_171 = Decimal(1), Decimal(1)
for n in range(1, 171): f_170 *= n
for n in range(1, 172): f_171 *= n
print(f'{f_171 - f_170:.5e}')
```

```
1.23376e+309
```

(e)

[5]:
```
a, b = np.double(1.000000000001), np.double(1.0)
direct, conjugate = np.sqrt(a) - np.sqrt(b), (a-b)/(np.sqrt(a) + np.sqrt(b))
d = 1e-12 / (np.sqrt(a) + np.sqrt(b))
print(direct, conjugate, d)
```

```
5.000444502911705e-13 5.000444502910455e-13 4.99999999999875e-13
```

The problem here is the possibility of catastrophic cancellation: when square rooting a and b, a is very close to machine precision. This will use up most of the accurate digits in the calculation, and the result will contain quite large relative numerical errors. Avoiding taking the square roots before subtraction will slightly improve accuracy, because of the number of digits needed to store $\sqrt{a}$ being larger. This should increase stability. But, since a - b is already very small, it experiences catastrophic cancellation as well. If the difference d between a and b is known, you can evaluate $\sqrt{a+d} - \sqrt{b}$ directly, which is accurate to the 11th digit, compared to 4 in the other methods (by Taylor exansion of $\sqrt{1+x}$, we know that the expression evaluates to 5e-13 + O(e-24)). This is a slight reformulation of the problem, but is a numerically stable approach for e.g. root finding functions where d is known ahead of time.

TFY4235: Computational Physics - 2D Ising Model Exam Report

Candidate ID: 10039

ABSTRACT

The two-dimensional Ising model serves as a fundamental tool in statistical physics for understanding phase transitions and collective behaviour in spin systems. This report presents a Monte Carlo simulation study of the model on square lattices using the Metropolis algorithm, with the aim of investigating thermodynamic properties near the critical temperature. The specific heat and magnetic susceptibility were computed to estimate the critical temperature and examine scaling relations. The impact of system size and impurity concentration on phase behaviour was investigated. Additionally, simulated annealing was employed to identify low energy spin configurations in the presence of quenched disorder. Results confirm expected phase transition features and demonstrate qualitative agreement with theoretical predictions, including the presence of domain structures, critical fluctuations, and susceptibility scaling. The estimated critical temperature for $L = 40$ deviates by less than 0.3% from the exact analytical value. Despite finite-size and sampling limitations, the numerical experiments effectively capture essential characteristics of second-order phase transitions.

I. INTRODUCTION

The Ising model is a fundamental model in statistical physics used to study phase transitions and collective behaviour in systems of interacting particles. In its two-dimensional form, it consists of spins arranged on a square lattice, where each spin can take values of +1 or –1 and interacts only with its nearest neighbours. Despite its simplicity, the model captures key features of real materials, including spontaneous magnetisation and critical phenomena near the phase transition point.[1]

The goal of this project is to simulate the two-dimensional Ising model using Monte Carlo methods in order to study its thermodynamic behaviour. Specifically, the critical temperature is estimated by analysing the specific heat and magnetic susceptibility of the system as functions of temperature. The influence of lattice size and the presence of impurity sites is also investigated. In addition, simulated annealing is applied to optimise spin configurations in the presence of impurities, and scaling relations are explored near the critical temperature.

The structure of the report is as follows: Section II presents the theoretical background of the Ising model and the algorithms used to solve it. Unless explicitly cited, the theoretical framework presented in this section is derived from the exam paper. Section III describes the numerical methods and simulation details. Section IV presents the results and discusses the key findings. Section V provides concluding remarks and perspectives for future work.

II. MODELS

A. The Ising Model

The two-dimensional Ising model describes a system of $N = L \times L$ discrete spin variables $s_i = \pm 1$, arranged on a square lattice. This model captures central features of phase transitions and critical behaviour with a simple two-state spin system. Its exact solution in two dimensions, ease of implementation, and suitability for numerical simulations make it a useful model in statistical physics.

The energy of the system is given by

$$E = -J \sum_{\langle i,j \rangle} s_i s_j - \mu H \sum_i s_i \tag{1}$$

where J is the coupling constant between nearest-neighbour spins, μ is the Bohr magneton, H is the external magnetic field, and $\langle i, j \rangle$ denotes summation over nearest-neighbour pairs.

The magnetisation of the system is given by

$$M = \sum_i s_i \tag{2}$$

The model exhibits a second-order phase transition at the critical temperature T_c, separating a low-temperature ordered (ferromagnetic) phase from a high-temperature disordered (paramagnetic) phase. The exact value of T_c for the 2D square lattice Ising model is given by[2]

$$T_c = \frac{2J}{k_B \ln(1+\sqrt{2})} \tag{3}$$

where k_b is the Boltzmann constant.

The specific heat is a measure of energy fluctuations in the system and is given by

$$C = \frac{(\Delta E)^2}{k_B T^2} \tag{4}$$

Near the critical temperature T_c, the specific heat diverges as a power law of the system,

$$C \sim |T - T_c|^{-\alpha} \tag{5}$$

were α is critical exponent which describes the phase transition.

The magnetic susceptibility, χ, is a measure of how much the magnetisation changes in response to an external magnetic field, and is given by

$$\chi = \frac{(\Delta M)^2}{k_B T} \tag{6}$$

Similarly to the specific heat, the magnetic susceptibility exhibits power law scaling near the critical temperature:

$$\chi \sim |T - T_c|^{-\gamma} \tag{7}$$

Near the critical point, the Ising model exhibits universal scaling behaviour. The magnetic susceptibility follows the scaling law

$$\chi(t,h) = |t|^{-\gamma} g_{\pm}\left(\frac{h}{|t|^{\beta\delta}}\right) \tag{8}$$

Here, $t \approx (T - T_c)/T_c$ and $h = \mu H/J$. $g_{\pm}$ are scaling functions above and below T_c. The critical exponents take values $\gamma = 7/4$, $\beta = 1/8$, and $\delta = 15$ for the 2D Ising model.

B. The Metropolis Algorithm

The Metropolis algorithm is a Markov Chain Monte Carlo (MCMC) method used to sample spin configurations according to the Boltzmann distribution. The probability of the system being in a given configuration with energy E_n is proportional to the Boltzmann factor

$$P(E_n) \sim e^{-E_n/k_B T} \tag{9}$$

In the Metropolis algorithm, spin flips are proposed at random and accepted with a probability A that depends on the resulting energy change,

$$A = \min\left(1, e^{-\Delta E/k_B T}\right) \tag{10}$$

Here, ΔE is the difference in energy from a proposed spin flip. The acceptance probability A is compared to a randomly drawn number r from a uniform distribution $r \in U(0,1)$. The proposed spin flip is accepted if $A > r$, and rejected otherwise. This ensures that energetically favourable moves ($\Delta E < 0$) are always accepted, while unfavourable moves are accepted with a probability that decreases exponentially with ΔE.

The acceptance condition makes the Metropolis algorithm more efficient than methods such as naive sampling. In naive sampling, a large percentage of randomly generated configurations have high energy and negligible probability, leading to slow convergence. The Metropolis algorithm improves efficiency by biasing the sampling toward low energy, high probability states, ensuring that most configurations contribute meaningfully to the estimation of statistical quantities.[3]

C. Simulated Annealing

Simulated annealing is a minimisation technique inspired by the physical process of slowly cooling a material to reach a low energy state. By gradually lowering the temperature during a simulation, the system is able to explore the energy landscape and increasingly favour lower energy configurations.[4] This method is easily combined with the Metropolis algorithm, and typically requires a relative low amount of sampling to find the low energy configurations.

III. METHODS

The two-dimensional Ising model was implemented on a square lattice of size $L \times L$ in Python, using periodic boundary conditions to eliminate edge effects. All simulations were carried out in units where the Boltzmann constant k_B, the coupling constant J, and the Bohr magneton μ were set to 1, effectively normalising the thermodynamic variables.

A. Random sampling

All random sampling was performed using `random` from the NumPy library[5], which employs a Piecewise Congruential Generator (PCG) to generate pseudo-random numbers. The PCG algorithm enhances quality in generated numbers by combining a linear congruential generator with an output permutation, offering improved statistical properties over basic linear congruential methods.[6]

Using pseudo-random numbers in algorithm development is practical, as it enables simulations to be seeded, ensuring reproducibility. However, the statistical quality depends on the random number generator, and once seeded, the process is fully deterministic.

B. Monte Carlo Simulations

Monte Carlo simulations of the Ising model were performed using the Metropolis algorithm described in Section II B. Each simulation started with a randomly initialised spin configuration on an $L \times L$ lattice, where each site was assigned a spin value of ± 1. At a given temperature T, spin flips were proposed randomly, and the resulting energy change, ΔE, was computed using Eq. 1. This calculation was done by evaluating ΔE locally, involving only the spin interactions with the four nearest neighbours. The acceptance probability A was determined using the computed energy change ΔE. A randomly drawn number $r \in U(0,1)$ was then compared to this probability. If $A > r$, the proposed spin flip was accepted. The energy E and magnetisation M of the system were then updated using the calculated energy change ΔE and the change in magnetisation due to the successful flip (Eq. 2). Spin flips were attempted N times, with each spin in the lattice averaging one attempted flip. Completing this set of N flip attempts is referred to as one *sweep* through the system.

The magnetisation M and energy E as functions of temperature were estimated by performing a fixed number of sweeps at each temperature T and calculating their averages afterward. Before the averages were computed, the system was allowed to equilibrate by doing a set amount of initial sweeps. This process was repeated across a predetermined set of temperatures to obtain the desired relations.

The critical temperature was estimated by calculating the specific heat using Eq. 4. For each fixed temperature T, the system was first equilibrated, after which a series of sweeps were performed to sample the energy E. The variance of these samples was then used to compute C at that temperature. As in previous procedures, the temperature was varied over a predefined range. Based on Eq. 5, where C diverges at T_c, the critical

temperature was identified as the value of T corresponding to the maximum of C, $T_c = T(C_{\max})$.

The magnetic susceptibility was computed using the same approach as for the specific heat, based on Eq. 6. For each temperature, the system was equilibrated and sampled to obtain the variance in magnetisation. Additionally, the strength of the external field H was varied to observe how χ responds to different field magnitudes.

Impurities were introduced into the system to study how various parameters respond to different concentrations of frozen spins. A predefined impurity fraction p, representing the percentage of total spins, was randomly selected within the lattice. This was implemented by generating a boolean mask array prior to initialisation, where the impurity sites were marked as true. During the Metropolis algorithm, any spin selected for a potential flip was checked against this mask, and if marked as an impurity, the spin was held fixed regardless of whether the flip condition was satisfied. Impurities were held static for each simulation duration to mimic the response of a real system at different temperatures.

Simulated annealing was performed by first introducing impurities, and then equilibrating the system at a high temperature. The temperature was then gradually decreased, with a number of sweeps carried out at each step. Unlike the previously described methods which involve averaging variables at each temperature, simulated annealing follows the evolution of the system across the temperature range, and only the end configuration is of interest.

To examine the scaling relation in Eq. 7, values of χ were computed for temperatures close to T_c. The rescaled variables $\chi|t|^{\gamma}$ and $h/|t|^{\beta\delta}$ were then plotted against each other to visually separate the two branches of the scaling function $g_{\pm}$. The theoretical value of T_c, given by Eq. 3, was used as the reference point for the scaling analysis.

For consistency across different system sizes, all variables were normalised by the system size $L \times L = N$ after computations. Throughout the rest of this report, these normalised quantities will be referred to as e.g. M and E for simplicity, although they technically represent M/N and E/N, respectively.

IV. RESULTS AND DISCUSSION

The results were obtained from simulations performed as described in Section III. Unless stated otherwise, the lattice size was fixed at $L = 40$, and all variables were averaged over 10 000 sweeps following 2 000 initial sweeps for equilibration. The results are divided into three sections: Section IV A examines key system variables in the absence of a magnetic field, Section IV B analyses how these variables are affected by the introduction of a magnetic field and Section IV C assesses the statistical reliability of the results based on convergence behaviour.

A. Task 3.1

A rough indication of the equilibration time can be drawn from Fig. 1. The system appears to reach equilibrium relatively quickly at $T = 2.1$ and $T = 2.5$, in under 1000 sweeps. In contrast, at $T = 2.3$, the system remains unstable throughout the duration of the simulation. A more rigorous assessment of the equilibration time would involve analysing the correlation time to determine when the system becomes statistically independent of its initial state. This is discussed further in Section IV C.

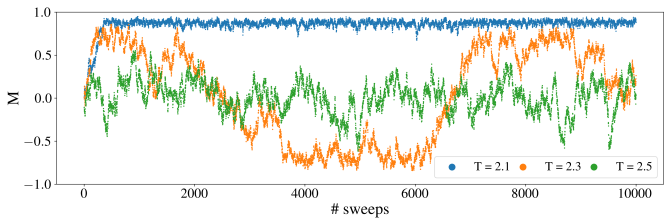

FIG. 1. Time evolution of the magnetisation (with sign) as a function of sweeps for temperatures $T = 2.1$, 2.3, and 2.5 on a 40×40 lattice with $H = 0$. The plot illustrates the equilibration behaviour and temperature-dependent fluctuations near the critical point.

Fig. 1 also shows how temperature influences the magnetisation of the spin system. As dictated by the sign of the coupling term in Eq. 1, neighbouring spins with opposite orientations are energetically unfavourable. In the absence of an external magnetic field, the ground state corresponds to all spins aligned, resulting in a normalised energy of $E = -2$ and magnetisation $M = \pm 1$. At higher temperatures, the Boltzmann factor $e^{-\Delta E/k_b T}$ increases for $\Delta E > 0$, increasing the probability that a flip is accepted, even if it is energetically unfavourable. This makes it more difficult for the system to stabilise in the lowest energy spin configurations.

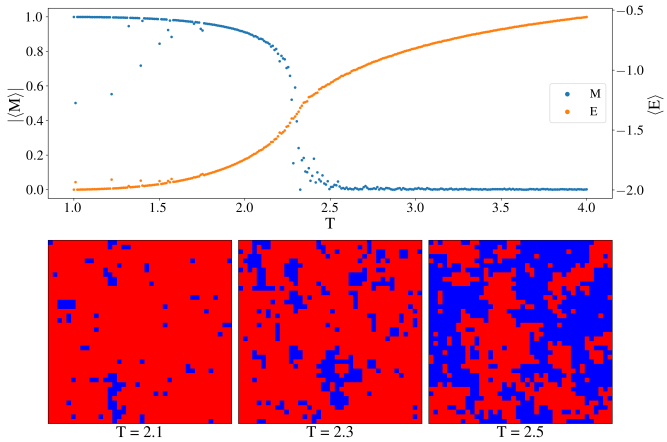

FIG. 2. Average absolute magnetisation and energy as functions of temperature for a 40×40 lattice with $H = 0$. Snapshots show representative spin configurations at three selected temperatures, illustrating domain formation and the phase transition.

The patterns observed in Fig. 1 can be examined further by computing averages of magnetisation and energy across a range of temperatures. As shown in Fig. 2, at low temperatures the system tends to remain in its lowest energy configurations, where all spins are aligned. As the temperature increases and approaches the critical temperature T_c, the system progressively becomes more disordered. However, for $T > T_c$, as illustrated for $T = 2.5$, the system in not entirely disordered. Instead, distinct magnetic domains begin to appear. This occurs because such domain structures are more energetically favourable

than full disorder, allowing the system to retain low energy despite an increase in misaligned spins. This behaviour resembles the properties of real ferromagnetic materials, which behave as magnets below T_c, and form larger domains with low net magnetism above T_c, becoming paramagnetic.[7]

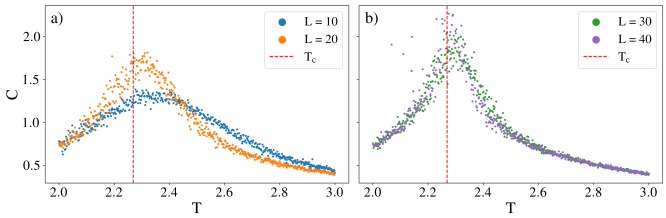

FIG. 3. Specific heat as a function of temperature for different lattice sizes. The theoretical critical temperature $T_c \approx 2.27$ is marked. The plots show the sharpening of the specific heat peak with increasing system size, reflecting finite-size effects.

The specific heat for different system sizes is shown in Fig. 3, where the expected divergence predicted by Eq. 5 is clearly visible. The trend from the different lattice sizes shows this divergence to be less prominent as lattice size decreases. This is expected, as fewer total spins means the variance in energy $(\Delta E)^2$ is naturally lower. Accurately capturing the divergence behaviour required to estimate the critical exponent α would therefore benefit from larger system sizes. In addition, finite-size scaling techniques would have to be applied to approximate the behaviour of infinite systems.[1] Given the methods outlined in Section III, obtaining a reliable estimate of α is unlikely.

L	10	20	30	40
$\mathbf{T_c}$	2.277 (0.33%)	2.321 (2.27%)	2.267 (0.12%)	2.275 (0.24%)

TABLE I. Estimated critical temperatures for different lattice sizes. Error, given in parentheses, is relative to theoretical T_c.

Despite the small system sizes, the critical temperature T_c has been estimated with reasonable accuracy, as shown in Tab. I. When compared to Fig. 3, a reduction in system size appears to shift the estimated T_c to higher temperatures. This trend is more apparent in the plotted data than in the estimates of T_c, likely due to statistical fluctuations. A more detailed discussion of this point is provided in Section IV C.

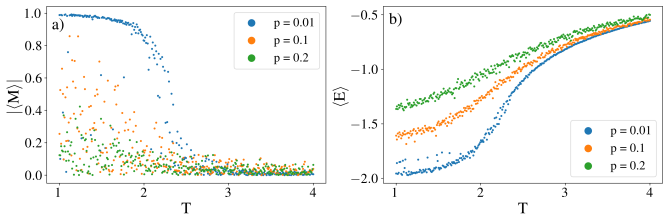

FIG. 4. System response to impurity concentrations $p = 0.01$, 0.10, and 0.20: (a) Average absolute magnetisation and (b) average energy as functions of temperature.

The introduction of impurity sites seems to inhibit ferromagnetic properties of the system at low temperatures, as seen in Fig. 4. From (a), it is clear that the system is less magnetised at low T. This is due to frozen spins not aligning. As a consequence, the total energy of the system is also higher at low T, as seen in (b).

In addition, the phase transition is observed to become increasingly blurred as the impurity concentration p rises. As shown in Fig. 5, the introduction of impurities appears to smooth out the divergence in C. This is analogous to reducing the system size (Fig. 3): with fewer spins available to flip, the fluctuations naturally become smaller. But unlike a smaller system, impurities cause more disorder, which shifts T_c to a lower value.

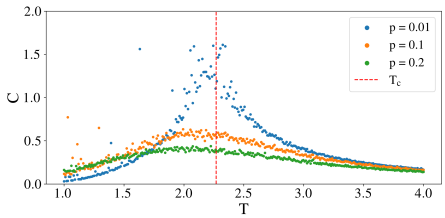

FIG. 5. Specific heat as a function of temperature for different impurity level. Increasing impurity concentration seemingly suppresses the specific heat peak and shifts the apparent critical temperature downward.

The simulated annealing results are presented in Tab. II. For each run, the temperature was gradually decreased from 5 to 0.1 across 1000 temperature steps, with 1000 sweeps performed at each temperature. The placement of impurities was kept static between runs to analyse the ability of simulated annealing to find the lowest energy state.

p	**Run 1**	**Run 2**	**Run 3**	$\langle \mathbf{E} \rangle \,\|\|\, \langle \mathbf{M} \rangle$
0.01	−1.965 || 0.991	−1.965 || 0.991	−1.970 || 0.993	−1.967 || 0.992
0.10	−1.640 || 0.739	−1.615 || 0.136	−1.657 || 0.441	−1.637 || 0.439
0.20	−1.395 || 0.076	−1.395 || 0.381	−1.397 || 0.165	−1.395 || 0.207

TABLE II. Simulated Annealing results. Final energies (E) and magnetizations (M) for three independent runs at each impurity concentration p. The first value in each entry corresponds to E, the second to M.

The energy values obtained across all runs are relatively consistent, suggesting that the simulated annealing process effectively succeeded in locating low energy states. However, the magnetisation varied between runs, indicating that distinct low energy configurations were reached in each case. This implies the presence of multiple low energy states with comparable energies, and that the simulated annealing implementation has limited ability to distinguish between them.

Fig. 6 indicates that as p increases, domain structures begin to form around impurities with matching spin orientations, suggesting that the impurities influence the magnetic domain formation. Although the effect is random in this case, it serves as a rudimentary illustration of how the magnetic properties of a material can be altered through deliberate introduction of impurities.

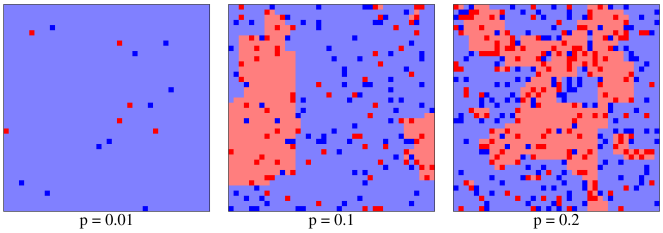

FIG. 6. Snapshots of the lowest-energy spin configurations obtained via simulated annealing for impurity levels $p = 0.01$, 0.10, and 0.20. Frozen impurity spins are highlighted. The results show domain structures forming around impurity patterns.

B. Task 3.2

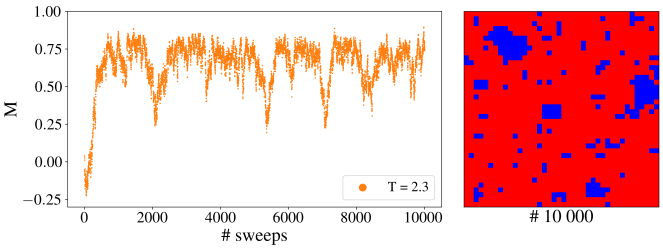

FIG. 7. Time evolution of magnetisation with sign for $T = 2.3$ and external field $H = 0.05$. A snapshot of the final spin configuration is included, showing field-aligned magnetisation with small opposing domains.

Fig. 7 shows the effect of an external magnetic field applied to the system, and is directly comparable to the zero-field case in Fig. 1 at the same temperature. With the external field applied, the magnetisation of the system aligns with the field, resulting in significantly reduced fluctuations in M. This behaviour follows from Eq. 1: the energy of the system is minimised when H and M are aligned. At this temperature and field strength, some spins may orient against the magnetic field, sometimes stabilised by forming small domains, as seen in the snapshot in Fig. 7. However, these domains remain too limited in size to reverse the overall direction of the magnetisation of the system.

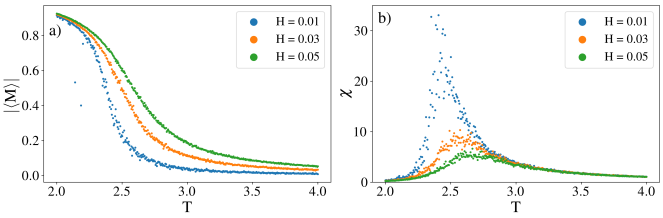

FIG. 8. (a) Average absolute magnetisation and (b) magnetic susceptibility as functions of temperature for various external field strengths $H = 0.01$ to 0.05. Stronger fields increase magnetisation and suppress susceptibility fluctuations near T_c.

The influence of an external magnetic field becomes especially clear when examining system behaviour across different temperatures. As illustrated in Fig. 8, stronger external fields lead to increased magnetisation and a suppression of fluctuations near the critical temperature, thereby reducing $(\Delta M)^2$. This effect results in a smoother phase transition. As the field strength increases, the system tends to maintain ferromagnetic order at higher temperatures. This demonstrates how an external magnetic field can be used effectively to induce magnetisation in ferromagnetic materials.

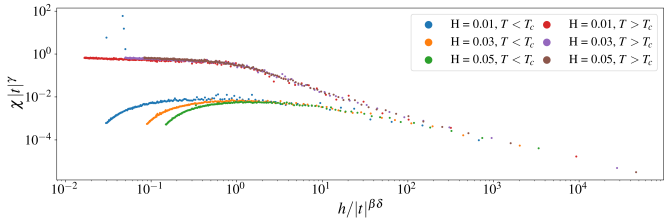

FIG. 9. Scaling collapse of the magnetic susceptibility χ near the critical temperature T_c. The rescaled variables $\chi|t|^{\gamma}$ and $h/|t|^{\beta\delta}$ are plotted against each other, where $t = (T - T_c)/T_c$ and $h = H$. The collapse onto two distinct branches indicate correct scaling behaviour near the phase transition.

Fig. 9 presents the scaling collapse of χ near T_c, based on data collected across a temperature range from 1 to 4 with 400 evenly spaced samples. Given the definition of t, the collapse is expected in the limit where $h/|t|^{\beta\delta}$ becomes large. For all values of H near T_c, the data appears to collapse onto a single curve. However, the relatively low data density in the vicinity of T_c limits the precision of the analysis, and higher resolution in this region would be beneficial. The two distinct branches of $g_{\pm}$ become apparent as T deviates from T_c, supporting that χ follows the expected scaling behaviour near the critical temperature. The branch splitting observed for $T < T_c$ is accounted for by the differing scaling behaviour of χ across various values of H, as observed in Fig. 8.

C. Convergence analysis

An essential feature of Monte Carlo simulations is the number of samples used in the analysis. Since the quantities of interest are probabilistic in nature, the accuracy of the simulation results depends significantly on the sample size. Examining how the results converge as the number of increases is crucial for assessing their reliability.

The results clearly show that statistical fluctuations are most prominent near the critical temperature T_c, as illustrated in Figs. 3, 5, and 8, among others. Therefore, analysing the convergence of variables is particularly crucial in this region.

Variable	10^4 sweeps	10^5 sweeps	10^6 sweeps	10^7 sweeps
$\langle M \rangle$	-0.0998	0.312	0.0314	0.0147
$(\Delta M)^2$	0.374	0.330	0.430	0.435
$\langle E \rangle$	-1.416	-1.427	-1.429	-1.430
$(\Delta E)^2$	0.00590	0.00641	0.00636	0.00628

TABLE III. Convergence analysis with all variables evaluated at the theoretical transition temperature T_c.

Tab. III shows how the variables at T_c depend on the number

of sweeps performed. As mentioned earlier, 10^4 sweeps were used to obtain the presented results. Near T_c, M exhibits rapid fluctuations, as illustrated in Fig. 1, which slows down the convergence of $\langle M \rangle$, a trend clearly reflected in the data. This also impacts the convergence of $(\Delta M)^2$, which differs by 14% when comparing 10^4 and 10^7 sweeps. The energy related quantities are more stable and show less dramatic variation, and the value of T_c in Tab. I is within 0.24% of the exact result. However, this estimate may not be reliably reproduced with the same simulation parameters, as $(\Delta E)^2$ changes by 6% between 10^4 and 10^7 sweeps. This indicates a considerable statistical uncertainty in T_c when using only 10^4 sweeps.

Slow convergence is a well known limitation of local update algorithms such as the Metropolis method when applied to the Ising model. The correlation time, τ, representing the number of sweeps required for the system to equilibrate, diverges as $\tau \sim L^z$ with $z \geq 2.0$.[8] Therefore, using 2000 sweeps for equilibration is a reasonable choice. However, to ensure statistical reliability of the simulation results, the number of sampling sweeps should be substantially larger than τ. The convergence analysis indicates that a precise characterisation of scaling behaviour near the critical threshold would thus require a substantially larger sample size than what is used in this work.

Despite certain limitations, the algorithm developed in this work demonstrates notable strengths. It is both conceptually straightforward and computationally efficient. With modest computational resources, it enables qualitative investigation of key physical phenomena, including phase transitions, domain configurations, and the ferromagnetic response to impurities and external magnetic fields. These simulated behaviours exhibit qualitative agreement with corresponding real world phenomena.

V. CONCLUSIONS

The Metropolis algorithm was employed to capture critical phenomena in the two-dimensional Ising model, providing estimates of the critical temperature close to the analytical result $T_c \approx 2.27$. Specific heat and magnetic susceptibility exhibited characteristic divergences near T_c, affirming the power-law behaviour typical of second-order phase transitions. The introduction of impurity sites significantly reduced magnetisation and elevated system energy, illustrating their role in altering magnetic domain formation and critical behaviour. Simulated annealing demonstrated multiple energetically comparable configurations, reflecting complexities inherent in real ferromagnetic systems. Although computational constraints limited precision in critical exponent estimates, the implemented methods captured qualitative trends, validating the Metropolis algorithm and simulated annealing as effective approaches in computational physics. To further improve the analysis, more advanced sampling methods such as Wolff or Swendsen-Wang cluster algorithms could be used to reduce critical slowing down and enhance convergence near T_c.

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VII. AI DECLARATION

Artificial Intelligence tools were used during the preparation of this work, strictly as a writing and debugging aid. All concepts, methodologies, analyses, and conclusions presented in this submission are entirely my own.

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